1.1 Abstract and concrete categories

Dfn. A **category** consists of

- a collection of **objects** $X, Y, Z, \ldots$
- a collection of **morphisms** $f, g, h, \ldots$

So that:

- Each morphism has specified **domain** and **codomain** objects; the notation $f : X \to Y$ signifies that f is a morphism with domain X and codomain Y
- Each object has a designated **identity morphism** $1_X : X \to X$
- For any pair of morphism f, g with the codomain of f equal to the domain of g, there exists a specified **composite morphism** gf whose domain is equal to the domain of f and whose codomain is equal to the codomain of g, i.e.,:

$$f : X \to Y, \qquad g : Y \to Z \quad \rightsquigarrow \quad gf : X \to Z$$

This data is subject to the following two axioms:

- For any $f : X \to Y$, the composites $1_Y f$ and $f 1_X$ are both equal to f.
- For any composable triple of morphisms f, g, h, the composites $h(gf)$ and $(hg)f$ are equal and henceforth denoted by hgf

$$f : X \to Y, \quad g : Y \to Z, \quad h : Z \to W \quad \rightsquigarrow \quad hgf : X \to W$$

  That is, the composition law is associative and unital with the identity morphisms serving as two-sided identites.

Dfn. A category is **concrete** when its objects are actually described as sets and its morphisms are actually functions between them and we refer to such morphisms as maps. Otherwise, the category is **abstract** and we refer to the morphisms as arrows

Dfn. A category is **small** if it has only a set's worth of morphisms.

Dfn. A category is **locally small** if between any pair of objects there is only a set's worth of morphisms. Then we *are justified* in writing:

$$\mathsf{C}(X, Y) \quad \text{or} \quad \mathrm{Hom}\,(X, Y)$$

and we call it a **hom-set**, but we *abusively* use the same notation for a **hom-class**.

Dfn. An **isomorphism** in a category is a morphism $f : X \to Y$ for which there exists a morphism $g : Y \to X$ so that $gf = 1_X$ and $fg = 1_Y$. The objects X and Y are **isomorphic** whenever there exists an isomorphism between X and Y, in which case one writes $X \cong Y$.

An **endomorphism** is a morphism with equal domain and codomain

An **automorphism** is an isomorphism that is an endomorphism

Dfn. A **groupoid** is a category in which every morphism is an isomorphism.

Dfn. A **subcategory** D of a category C is a restriction of a category to a subcollection of objects and a subcollection of morphisms such that

(i) The domains and codomains of morphisms in D are contained in D

(*ii*) The identities of all objects in D are contained in D

(*iii*) The compositions of every composable pair of morphisms in D are contained in D

Lemma. Any category C contains a **maximal groupoid**, the subcategory containing all of the objects and only those morphisms that are isomorphisms.

Exercise (1.1.i)**.**

(*i*) consider a morphism $f : x \to y$. Show that if there exists a pair of morphisms $g, h : y \rightrightarrows x$ so that $gf = 1_x$ and $fh = 1_y$, then $g = h$ and f is an isomorphism.

(*ii*) Show that a morphism can have at most one inverse isomorphism.

<u>*Proof*</u> (i)

Assume $\exists g, h : y \rightrightarrows x$ so that $gf = 1_x$ and $fh = 1_y$. Then

$$g \underset{\text{morphism dfn.}}{=} g1_y \underset{\text{asumptions}}{=} g(fh) \underset{\text{associativity}}{=} (gf)h \underset{\text{assumptions}}{=} 1_x h \underset{\text{identity}}{=} h \tag{$\star$}$$

□

<u>*Proof*</u> (ii)

Let $g, h : y \rightrightarrows x$ be inverse isomorphisms of $f : x \to y$

But this implies $gf = 1_x$ and $fh = 1_y$

Hence by the argument presented in ($\star$) we obtain $g = h$.

□

Exercise (1.1.ii)**.** Let C be a category. Show that the collection of isomorphisms in C defines a subcategory, the **maximal groupoid** inside C

<u>*Proof*</u>

Let C be a category

Consider the following data D (we haven't yet confirmed it is a category):

$$\begin{aligned} &\mathrm{Obj}\,(\mathsf{D}) := \mathrm{Obj}\,(\mathsf{C}) \\ &\forall x, y \in \mathrm{Obj}\,(\mathsf{D}) \implies \mathsf{D}\,(x, y) := \{f \in \mathsf{C}\,(x, y) \mid f \text{ is an isomorphism}\} \end{aligned}$$

It suffices to verify the 3 subcategory conditions for D:

(i) No object has been removed, so domains and codomains of all *iso*morphisms are still present

(ii) Identities are isomorphisms, so remain in our definition of D

(iii) The composite of composible pairs of isomorphisms are isomorphisms that remain in D:

$$(f^{-1}g^{-1})(gf) = 1_X$$

□

Exercise (1.1.iii.(i))**.** For any category C and any object $c \in \mathsf{C}$ show that there is a category c/C whose objects are morphisms $f : c \to x$ with domain c and in which a morphism from $f : c \to x$ to $g : c \to y$ is a map $h : x \to y$ between the codomains so that the triangle

$$\begin{tikzcd} & c \arrow[dl, "f"'] \arrow[dr, "g"] & \\ x \arrow[rr, "h"'] & & y \end{tikzcd}$$

commutes (i.e. so that $g = hf$)

<u>*Proof*</u> (i)

Let C be a category with $c \in \mathrm{Obj}\,(\mathsf{C})$

Consider the following data c/C (we haven't yet confirmed it is a category):

$\mathrm{Obj}\,(c/\mathsf{C}) := \{f : c \to x \mid x \in \mathrm{Obj}(\mathsf{C})\}$
$\forall^{f:c\to x}_{g:c\to y} \in \mathrm{Obj}\,(c/\mathsf{C}) \implies \mathrm{Hom}_{c/\mathsf{C}}\,(f, g) := \{h : x \to y \in \mathsf{C} \mid g = h \circ f\}$
it suffices to prove that $\mathrm{Mor}\,(c/\mathsf{C})$ is closed under composition and contains identities, but this is true since:

left composition with 1_f gives the identity of $f \in \mathrm{Mor}\,(\mathsf{D})$

for all $f, g \in \mathrm{Mor}\,(D)$ we can define fg as $f \circ g\circ$

transitivity must hold since it holds for C

$\square$

Exercise (1.1.iii.(ii))**.** For any category C and any object $c \in \mathsf{C}$ show that there is a category C/c whose objects are morphisms $f : x \to c$ with domain c and in which a morphism from $f : x \to c$ to $g : y \to c$ is a map $h : x \to y$ between the domains so that $f = gh$.

<u>*Proof*</u> (ii)

Essentially the same arguement but using right composition

$\square$

1.2. Duality

Dfn. Let C be a category. The **opposite category** C^{op} consists of

- The objects of C
- For each morphisms $f : X \to Y$ in C a morphism $f^{\mathsf{op}} : Y \to X$ with domain and codomain swapped.

Dfn (Post/Pre-composition)**.** Fix C with $f : x \to y$ a morphism in C.

(i) define the post-composition of f:

$$\forall c \in \mathsf{C} \quad f_* : \mathsf{C}(c, x) \to \mathsf{C}(c, y) \text{ by the rule } f_*(u) := u \mapsto f \circ u$$

(ii) and define the pre-composition of f:

$$\forall c \in \mathsf{C} \quad f^* : \mathsf{C}(y, c) \to \mathsf{C}(x, c) \text{ by the rule } f_*(v) := v \mapsto v \circ f$$

Lemma (Lemma 1.2.3.i)**.** $\forall c \in \mathsf{C}$

$$f : x \to y \text{ is an isomorphism in } \mathsf{C} \iff f_* \text{ is an isomorphism in } \mathsf{Set}$$

Remark. *put plainly we are saying that if there is an isomorphism between two objects in a category then if we consider the* <u>*pair of sets*</u> *of morphisms* ***to*** *(and* ***from****) the two objects to any particular fixed object then those morphisms can be put into a bijection.*

<u>*Proof*</u> $(\Rightarrow)$

Assume $f : x \to y$ is an isomorphism in C.

Then, it has inverse $g : y \to x$.

Consider the post-composition of $u \in \mathsf{C}(c, y)$ with g:

$$g_* : \mathsf{C}(c, y) \to \mathsf{C}(c, x) \qquad g_* := u \mapsto g \circ u$$

Take $h \in \mathsf{C}(c, x)$

$$\begin{aligned}
g_* f_*(h) &= g_*(f(h)) && \forall f \in \mathsf{C}(x, y) && && \text{dfn. post-comp.} \\
&= g(f(h)) && \forall f \in \mathsf{C}(x, y) && \forall g \in \mathsf{C}(y, x) && \text{dfn. post-comp.} \\
&= (gf)h && \forall f \in \mathsf{C}(x, y) && \forall g \in \mathsf{C}(y, x) && \text{assoc} \\
&= 1_x h && && && \text{dfn. of inv. func.} \\
&= h && && && \text{identity}
\end{aligned}$$

since this is true for arbitrary $h \in \mathsf{C}(c, x)$ we see $g_* f_* = 1_{\mathsf{C}(c,x)}$

Take $k \in \mathsf{C}(c, y)$

$$\begin{aligned}
f_* g_*(k) &= f_* g(k) && \forall g \in \mathsf{C}(y, x) && && \text{dfn. post-comp.} \\
&= f(g(k)) && \forall g \in \mathsf{C}(y, x) && \forall f \in \mathsf{C}(x, y) && \text{dfn. post-comp.} \\
&= (fg)k && \forall g \in \mathsf{C}(y, x) && \forall f \in \mathsf{C}(x, y) && \text{assoc} \\
&= 1_y k && && && \text{dfn. of inv. func.} \\
&= k && && && \text{identity}
\end{aligned}$$

since this is true for arbitrary $k \in \mathsf{C}(c, y)$ we see $f_* g_* = 1_{\mathsf{C}(c,y)}$

Hence, since $g_* f_* = 1_{\mathsf{C}(c,x)}$ and $f_* g_* = 1_{\mathsf{C}(c,y)}$ we see that g_* is the inverse morphism of f_* in the category of Set

$\square$

Proof ($\Leftarrow$)

Assume $f_* : \mathsf{C}(c,x) \to \mathsf{C}(c,y)$ is an isomorphism in Set and call its inverse morphism $\mathfrak{g}$:

$$\mathfrak{g} f_* = 1_{\mathsf{C}(c,x)} \text{ and } f_* \mathfrak{g} = 1_{\mathsf{C}(c,y)}$$

MUST COMEBACK AND FINISH

$\square$

Lemma (1.2.3.ii)**.**

$$f_* \text{ has an inverse} \leftrightsquigarrow f^* \text{ has an inverse}$$

Proof

MUST COMEBACK AND FINISH

$\square$

Dfn. Let C be a category with morphism $f : x \to y$. Then f is...

(i) a **monomorphism** (or just a **mono**) if for any parallel morphisms $h, k : w \rightrightarrows x$

$$fh = fk \text{ implies that } h = k$$

and we say f is **monic**

(ii) a **epimorphism** (or just a **epi**) if for any parallel morphisms $h, k : y \rightrightarrows z$

$$hf = kf \text{ implies that } h = k$$

and we say f is **epic**

Dfn (alt)**.** Let C be a category with morphism $f : x \to y$. Then f is...

(i) a **monomorphism** iff post-composition (f_*) is an injection on .

(ii) a **epimorphism** iff pre-composition (f_*) is injective

mono	$\forall h, k : w \rightrightarrows x \quad fh = fk \rightsquigarrow h = k$	$\rightarrowtail$
epi	$\forall h, k : y \rightrightarrows z \quad hf = kf \rightsquigarrow h = k$	$\twoheadrightarrow$

Dfn. suppose $x \xrightarrow{s} y \xrightarrow{r} x$ s.t $rs = 1_x$

- s is a **section** or **right inverse** to r
- r is a **retraction** or **left inverse** to s
- and we say s and r express the object x as a **retract** of the object y

Further, we call s a **split monomorphism** and r a **split epimorphism**

Lemma (1.2.11)**.**

- if f, g are mono ($\rightarrowtail$) then so is $g \circ f$
- if f, g are epi ($\twoheadrightarrow$) then so is $g \circ f$

contrapositively

- if $g \circ f$ is mono ($\rightarrowtail$) then so is f
- if $g \circ f$ is epi ($\twoheadrightarrow$) then so is g

Exercise (1.2.i)**.** Defining C/c to be $(c/(\mathsf{C}^{\mathrm{op}}))^{\mathrm{op}}$, deduce Exercise 1.1.iii(ii) from Exercise 1.1.iii(i)

Exercise (1.2.ii)**.** (*i*) Show that a morphism $f : x \to y$ is a split epimorphism in a category C if and only if for all $c \in \mathsf{C}$, the post composition function $f_* : \mathsf{C}(c, x) \to \mathsf{C}(c, y)$ is surjective.

(*ii*) Argue by duality that f is a split monomorphism if and only if for all $c \in \mathsf{C}$, the precomposition function $f^* : \mathsf{C}(y, c) \to \mathsf{C}(x, c)$ is surjective

Exercise (1.2.iii)**.** Prove Lemma 1.2.11 by proving either either epi or mono pairs of statements then using duality to prove the other pair. Prove that epi form a subcategory and dually mono

Exercise. iv, v, vi, vii

1.3 Functoriality

Dfn. A **functor** (covariant) $F : \mathsf{C} \to \mathsf{D}$, between categories C and D, consists of the following data:

An object $Fc \in \mathsf{D}$, for each object $c \in \mathsf{C}$

A morphism $Ff : Fc \to Fc' \in \mathsf{D}$, for each morphism $f : c \to c' \in \mathsf{C}$, so that the domain and codomain of Ff are, respectively, equal to F applied to the domain or codomain of f

The assignments are required to satisfy the following two **functoriality axioms**

For any composable pair $f, g \in \mathsf{C}$, $Fg \cdot Ff = F(g \cdot f)$

For each object $c \in \mathsf{C}$, $F(1_c) = 1_{F_c}$

Dfn. A **functor** (contravariant) F from C to D is a functor $F : \mathsf{C}^{op} \to \mathsf{D}$. Explicitly, this consists of the following data:

An object $Fc \in \mathsf{D}$, for each object $c \in \mathsf{C}$

A morphism $Ff : Fc' \to Fc \in \mathsf{D}$, for each morphism $f : c \to c' \in \mathsf{C}$, so that the domain and codomain of Ff are, respectively, equal to F applied to the codomain or domain of f

The assignments are required to satisfy the following two **functoriality axioms**

For any composable pair $f, g \in \mathsf{C}$, $Ff \cdot Fg = F(g \cdot f)$

For each object $c \in \mathsf{C}$, $F(1_c) = 1_{F_c}$

Remark. even in the presence of opposite categories, we always make an effort to draw arrows pointing in the "correct way" and depict composition in the usual order.

Lemma. Functors preserve isomorphisms. or

$$\text{if } f : c \to c' \text{ is an isomorphism } \underline{\text{then}}\ Ff : Fc \to Fc' \text{ is an isomorphism}$$

Proof

Consider a functor $F : \mathsf{C} \to \mathsf{D}$ and an isomorphism $f : x \to y$ in C with inverse $g : y \to x$.

Apply the two functoriality axioms:

$$F(g)F(f) = F(gf) = F(1_x) = 1_{F_x}$$

So, $fg : Fy \to Fx$ is a left inverse to $Ff : Fx \to Fy$.

Exchanging the roles of f and g shows that Fg is also a right inverse.

□

Dfn. If C is locally small, then for any object $c \in \mathsf{C}$ we define a pair of covariant and contravariant **functors represented by** c

$$\mathcal{C} \xrightarrow{\mathcal{C}(c,-)} \mathbf{Set} \qquad \mathcal{C}^{\mathrm{op}} \xrightarrow{\mathcal{C}(-,c)} \mathbf{Set}$$

$$\begin{tikzcd}
x \arrow[r, "f"] \arrow[d, mapsto] & y \arrow[d, mapsto] \\
\mathcal{C}(c,x) \arrow[r, "f_*"'] & \mathcal{C}(c,y)
\end{tikzcd}
\qquad
\begin{tikzcd}
x \arrow[r, "f"] \arrow[d, mapsto] & y \arrow[d, mapsto] \\
\mathcal{C}(x,c) & \mathcal{C}(y,c) \arrow[l, "f^*"']
\end{tikzcd}$$

Proposition. the covariant and contravariant **functors represented by** c satisfy the functoriality axioms

Proof (left to me) hint on pg 19

□

Dfn. a functor of two varibles is called a bifunctor

Dfn. For any categories C and D, there is a category $\mathsf{C} \times \mathsf{D}$, their **product**, whose

- objects are ordered pairs (c, d), where c is an object of C and d is an object of D
- morphisms are ordered pairs $(f, g) : (c, d) \to (c', d')$, where $f : c \to c' \in \mathsf{C}$ and $g : d \to d' \in \mathsf{D}$
- in which composition and identities are defined componentwise

Dfn. If C is locally small, then there is a **two-sided represented functor**

$$\mathsf{C}(-,-) : \mathsf{C}^{op} \times \mathsf{C} \to \mathsf{Set}$$

defined in the evident manner. A pair of objects (x, y) is mapped to the hom-set $\mathsf{C}(x, y)$. A pair of morphisms $f : w \to x$ and $h : y \to z$ is sent to the function

$$\mathsf{C}(x,y) \xrightarrow{f^*, h_*} \mathsf{C}(w,z)$$

$$g \longmapsto hgf$$

that takes an arrow $g : x \to y$ and then pre-composes with f and post-composes with h to obtain $hgf : w \to z$

1.4 Naturality

brief discursion on the linear dual

Dfn. Let V be a finite-dimension $\Bbbk$-vector space. A linear map $\psi : V \to \Bbbk$ is a **linear functional** on V

- the **linear dual** of V is given:

$$\begin{aligned} V^* :=& \operatorname{Hom}(V, \Bbbk) \\ =& \{\psi \in \mathcal{F}(V \to \Bbbk) \,:\, \psi \text{ is a linear map}\} \end{aligned}$$

- the **double dual** of V is given:

$$\begin{aligned} V^{**} :=& \operatorname{Hom}(\operatorname{Hom}(V, \Bbbk), \Bbbk) \\ =& \{\Psi \in \mathcal{F}(V^* \to \Bbbk) \,:\, \Psi \text{ is a linear map}\} \end{aligned}$$

Dfn. Given categories C and D and functors $F, G : \mathsf{C} \rightrightarrows \mathsf{D}$, a **natural transformation** $\alpha : F \Rightarrow G$ consists of:

- an arrow $\alpha_c : Fc \to Gc$ in D for each object $c \in \mathsf{C}$, the collection of which define the **components** of the natural transformation, so that for any morphism $f : c \to c'$ in C, the following square of morphisms in D:

$$\begin{tikzcd} Fc \arrow[r, "\alpha_c"] \arrow[d, "Ff"'] & Gc \arrow[d, "Gf"] \\ Fc' \arrow[r, "\alpha_{c'}"] & D \end{tikzcd}$$

commutes, i.e., has a common composite $Fc \to Gc'$ in D

- More succinctly we write describe just the *boundary data*:
- add globular diagram
- A **natural isomorphism** is a natural transformation $\alpha : F \implies G$ in which every component α_c is an isomorphism. In this case, the natural isomorphism may be depicted as $\alpha : F \cong G$